DD-Mamba: A Forecast-Decomposable State-Space Framework for Diagnosing When Time, Frequency, and Cross-Variable Components Help Multivariate Forecasting

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Abstract

From power grids to road networks, long-term multivariate forecasting must capture local time-domain dynamics, periodic structure that is compact in the frequency domain, and cross-channel dependencies whose strength varies enormously across datasets: the mean absolute inter-channel correlation spans 0.31 on ETTh1 to 0.91 on Solar. Combining time- and frequency-domain representations, and doing so with state-space backbones, is by now well explored. We therefore do not claim dual-domain or time–frequency Mamba modeling as novel; instead we ask *when* each component actually pays. We present DD-Mamba, a *forecast-decomposable* state-space framework built for that question: a decomposition-based time branch and a learned spectral branch each emit a full forecast, combined by a constrained convex gate, so either branch can be removed and its contribution measured directly; a DLinear-style backbone with zero-initialized corrections keeps the initial prediction linear in the instance-normalized window; and

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a bidirectional Mamba over variates is a per-dataset switch guided by measured channel coupling. On nine look-back-96 benchmarks DD-Mamba is competitive with results reported under the same protocol, with its clearest margins on Solar and ETTh1; because baselines are quoted rather than rerun in one pipeline, we make no state-of-the-art claim. The framework's purpose is diagnostic. Across 83 seed-paired comparisons with Benjamini–Hochberg control, all 29 component-placement effects are exploratory (none survive correction at n=3), while 9 of 54 branch-removal tests survive: the time branch is robustly necessary on the ETTm datasets, whereas confirmatory evidence for the frequency branch is limited to Solar at the 96-step horizon. DD-Mamba thus offers a reproducible way to diagnose conditional component effectiveness — which data characteristics reward which components — rather than another universally superior forecaster.
Keywords: Time series forecasting, State-space models, Mamba, Frequency domain, Multivariate analysis

1. Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [1, 2, 3], patch-based attention [4], inverted variate-token attention [5], and, most recently, selective state-space models (SSMs) built on Mamba [6, 7, 8], which offer linear-time sequence modeling with input-dependent dynamics.

Combining time- and frequency-domain representations is itself an established direction: TF4TF [9] models multi-semantic time–frequency information for long-term forecasting, and recent work pairs time–frequency or decomposition-based modeling with Mamba backbones [10, 8]. We therefore do *not* position dual-domain forecasting or a time–frequency Mamba as our contribution. What remains under-examined is a different question — *when* does each such component actually pay? Existing dual-domain models fuse representations inside the network, so the time view, the spectral view, normalization, and cross-variate capacity are entangled and cannot be individually switched off and measured. Our contribution is a *forecast-decomposable* framework that makes exactly these choices separable and testable, together with the seed-paired, multiplicity-controlled evidence of where each one helps.

Three empirical gaps motivate this design. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [11]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [12]) can be competitive with much larger models on standard benchmarks. Without such a direct pathway, a deep architecture must learn any linear behavior implicitly, which may be inefficient on small datasets. **Gap 2: the useful inductive bias differs by domain, but is rarely separable.** Local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain. Models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [13] — expose just one; and even dual-domain models that use both, such as TF4TF [9], fuse them *inside* the network, so neither view is a separately removable forecast path whose marginal value can be measured. **Gap 3:**

cross-variate capacity can help or hurt. Cross-variate mixing is effective on several high-dimensional benchmarks [5, 7, 14], but the underlying data property it exploits — how strongly channels are actually correlated — differs by a factor of three across the standard benchmarks (Table 1), and our controlled results show the same module can overfit a weakly coupled, low-channel dataset. This motivates treating the mixer as an explicit, data-diagnosable option rather than a universal setting.

We answer these three gaps with DD-Mamba (Fig. 1), mapping each to one structural remedy: *(i)* a DLinear-style time backbone with zero-initialized nonlinear corrections; *(ii)* parallel time and frequency forecasts, including an optional dormant FITS-style spectral map, fused by a convex gate; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Paired ablations test these choices below; only effects whose confidence intervals exclude zero are treated as supported.

Concretely, the time branch combines a DLinear-style pathway [11] with a zero-initialized nonlinear correction, while the frequency branch combines learned filtering with an optional zero-initialized FITS-style spectral pathway [12]. The resulting model starts as a linear map of the instance-normalized window (the RLinear-class structure [15, 16]); a controlled test finds zero initialization accuracy-neutral. The branch forecasts form a learned convex combination. Because each branch emits a full forecast, we can remove it and measure the effect with seed-paired tests, correcting for multiple comparisons with the Benjamini–Hochberg (BH) procedure. Two patterns emerge. Component-placement effects — the variate mixer helping Weather while hurting ETTh1, normalization helping Weather while hurting Solar — are visible at the raw level but, at three seeds, *none* survive BH correction; we therefore report them as *exploratory* directional evidence, not confirmed effects. Branch removal is sharper: of 54 branch tests, 9 survive BH, and they are dominated by the *time* branch (necessary on ETTm1 at every horizon and on ETTm2, Electricity, and Exchange at H=96). Confirmatory support for the *frequency* branch narrows after correction to a single cell, Solar at H=96; its raw significance on ETTm1, Electricity, Exchange, and ETTm2@336 does not survive BH. The dual-domain benefit is thus real but sharply conditional, and honestly characterized as exploratory beyond Solar@96.

Contributions. (1) A *forecast-decomposable* framework in which the time branch, spectral branch, normalization, and cross-variate mixer are each an independently removable component, so their marginal value can be measured directly rather than inferred from an end-to-end score; **(2)** a multiplicity-controlled diagnostic study — 83 seed-paired comparisons with Benjamini–Hochberg correction — that reports *which* components are confirmed (the time branch on the ETTm datasets; the frequency branch only on Solar@96) and which are merely exploratory (all component-placement effects at n=3), rather than overclaiming from raw p-values; **(3)** a concrete architecture realizing the framework: a DLinear-style time backbone with zero-initialized corrections (making the initial forecast an interpretable linear map, shown accuracy-neutral) and an optional dormant spectral pathway; and **(4)** a data-characteristic account — linking cross-channel PCC and horizon to component value — that turns the study into a practical, validation-only component-selection screen.

Organization. Section 2 reviews time series analysis tasks, deep models for temporal and for cross-dimensional dependencies, and the resulting research gap. Section 3 formalizes the forecasting problem and reviews the state-space background DD-Mamba builds on. Section 4 details the dual-domain architecture, its zero-initialized corrections, and the switchable variate mixer. Section 5 reports the main comparison, paired ablations, and the branch-ablation matrix; Section 6 distills practical component-selection guidance and threats to validity; and Section 7 concludes.

2. Related Work

2.1. Time Series Analysis Tasks

Deep models now serve the full spectrum of time series mining tasks — forecasting, classification, anomaly detection, and imputation — often through shared general-purpose backbones such as TimesNet [17], whose representations transfer across tasks. Long-term multivariate *forecasting*, our setting, is the most benchmark-standardized of these problems: fixed look-back, multi-horizon evaluation, and shared splits [1, 5] make it the natural arena for controlled architectural evidence, and findings about which structures a dataset rewards typically carry over to the sibling tasks that reuse forecasting backbones.

2.2. Temporal-Dependency Models

One line models dependencies along the time axis only. Transformer variants adapt attention to long sequences via sparsity, decomposition, and frequency enhancement [1, 2, 3], patching [4], or explicit de-/re-stationarization [18]. A second family shows that *linear* maps remain competitive: DLinear/NLinear [11], RLinear with reversible instance normalization [15, 16], the MLP-based TiDE [19] and TimeMixer [20], and the residual-stacked N-BEATS/N-HiTS [21, 22], our closest residual-forecasting precedents. A third works in the spectrum: FreTS [13] learns frequency-domain MLPs and FITS [12] forecasts with one complex-linear interpolation of the low-passed spectrum. A fourth, directly adjacent to us, combines *both* time and frequency views: TF4TF [9] mines multi-semantic time–frequency information for long-term forecasting. Such dual-domain models fuse the two views inside the network, so — unlike the framework here — neither view is a separately removable forecast path. Decomposition-based designs (STL [23], SCINet [24], and the seasonal-trend-dispersion view of STD [25]) make component structure explicit; RevIN, used here as a per-dataset switch, normalizes windows reversibly but neither extracts nor forecasts dispersion, and we claim no STD-style decomposition.

2.3. Cross-Dimensional (Spatial–Temporal) Models

A second line treats the variate dimension as a first-class axis. Crossformer [14] is the classic Transformer of this type: it embeds the series into a two-dimensional patch grid and attends over *both* the temporal and the cross-variate (spatial) dimension. iTransformer [5] inverts the axes and attends over variate tokens only; SOFTS [26] reaches linear-complexity channel interaction through a centralized core, and ModernTCN [27] interleaves temporal and cross-variable convolutions. Selective state-space models bring linear-time scans to the same axes: Mamba [6, 28] over time, and S-Mamba [7], TimeMachine [29], and FMamba [30] over variate tokens, with recent variants adding multi-scale processing (ms-Mamba [8]) and series decomposition (DecMamba [10]). DD-Mamba sits in this lineage — causal Mamba over time, bidirectional Mamba over variates and frequency bins — but, rather than proposing yet another fixed Mamba topology, treats the spatial dimension as a *switchable* module whose value is measured per dataset.

2.4. Research Gap

Dual-domain forecasting [9], time–frequency and decomposition Mamba hybrids [10, 8], and cross-dimensional attention [14, 5] are all, by now, well-populated spaces; a new model in any of them is unlikely to be a contribution on its own. What these lines share is a *methodological* limitation rather than an architectural one. (i) Representations are fused inside the network, so the marginal value of the time view, the spectral view, or a decomposition step is never separately measured. (ii) Cross-dimensional models such as Crossformer apply the same spatial–temporal capacity to every dataset, although channel coupling varies enormously across benchmarks (mean absolute cross-channel Pearson correlation ranges from 0.31 on ETTh1 to 0.91 on Solar; Table 1). (iii) Component-level comparisons, when reported at all, typically rest on a handful of seeds and uncorrected p-values, so it is unclear which claimed effects would survive multiplicity control. The gap we target is therefore diagnostic: no prior work, to our knowledge, pairs a per-component switchboard with controlled, multiplicity-corrected, seed-paired evidence of *where* — for which datasets, correlation regimes, and horizons — each structure pays. DD-Mamba is built to answer exactly that question: independently removable time/frequency forecast paths, a switchable variate mixer, zero-initialized corrections in the spirit of ReZero [31] (an interpretable initialization convention, shown accuracy-neutral), and a 27-cell branch-ablation matrix, analyzed under Benjamini–Hochberg control, that connects data characteristics to component value.

3. Preliminaries

3.1. Problem Formulation

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L\times C}$ of L steps and C variates, long-term multivariate forecasting predicts the next H steps $\mathbf{Y} \in \mathbb{R}^{H\times C}$, with H typically in $\{96, 192, 336, 720\}$. Following RevIN [16], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, using the window's own per-variate mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$, and de-normalize the final forecast by the inverse affine map. Instance normalization is treated as a per-dataset switch rather than a universal default; Section 5.3 shows why. A model is a map $f_\theta : \tilde{\mathbf{X}} \mapsto \mathbf{Y}$ trained to minimize the mean squared error over the horizon.

3.2. State Space Models

A continuous linear state-space model maps an input signal $u(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an N-dimensional latent state $\mathbf{h}(t) \in \mathbb{R}^N$,

$$\mathbf{h}'(t) = \mathbf{A}\,\mathbf{h}(t) + \mathbf{B}\,u(t), \qquad y(t) = \mathbf{C}^\top \mathbf{h}(t), \tag{1}$$

with $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B}, \mathbf{C} \in \mathbb{R}^N$. To operate on a discrete sequence, the system is discretized with a step size $\boldsymbol{\Delta}$ under a zero-order hold, which yields the discrete parameters

$$\bar{\mathbf{A}} = \exp(\boldsymbol{\Delta}\mathbf{A}), \qquad \bar{\mathbf{B}} = (\boldsymbol{\Delta}\mathbf{A})^{-1}(\exp(\boldsymbol{\Delta}\mathbf{A}) - \mathbf{I})\,\boldsymbol{\Delta}\mathbf{B} \;\approx\; \boldsymbol{\Delta}\mathbf{B}, \tag{2}$$

and turns Eq. (1) into the linear recurrence $\mathbf{h}_k = \bar{\mathbf{A}}\,\mathbf{h}_{k-1} + \bar{\mathbf{B}}\,u_k$, $y_k = \mathbf{C}^\top \mathbf{h}_k$. Classical structured SSMs keep $(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C})$ fixed across the sequence, so the

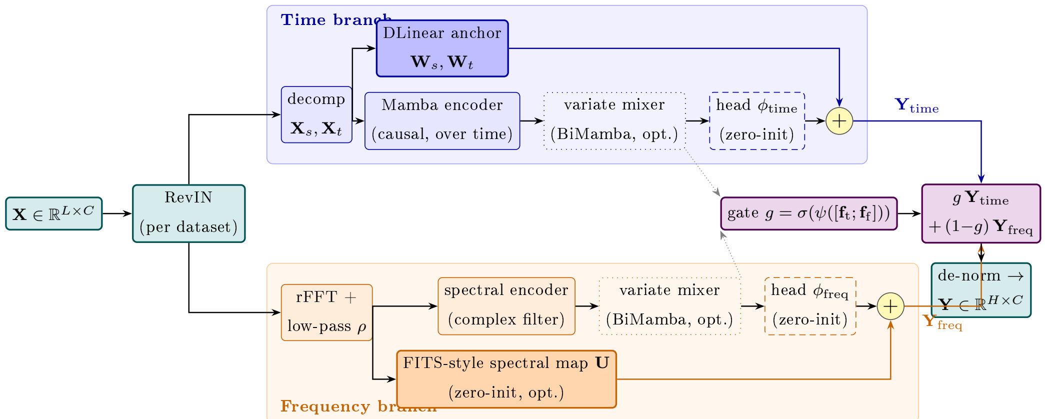

Figure 1: DD-Mamba overview, color-coded by role: the **normalization** wrapper (teal), the **time branch** (blue), the **frequency branch** (orange), and the **convex-gate fusion** (violet). In each time branch, a solid-filled DLinear-style backbone runs in parallel with a zero-initialized nonlinear head (dashed). The optional frequency-domain map and its nonlinear head are also zero-initialized, so the model begins as a scaled linear map of the instance-normalized window. Dotted boxes are per-dataset switches; dotted grey arrows carry branch features to the gate, whose convex weight g mixes the two coloured forecast paths $\mathbf{Y}_{\text{time}}$ and $\mathbf{Y}_{\text{freq}}$.

recurrence is linear time-invariant and can be evaluated as a global convolution. Mamba [6] makes the model *selective*: the parameters $(\boldsymbol{\Delta}_k, \mathbf{B}_k, \mathbf{C}_k)$ become functions of the input u_k, so the state transition adapts at each step and the model can selectively remember or forget context. Selectivity breaks the convolutional form, but a hardware-aware parallel scan keeps the cost linear in sequence length. DD-Mamba uses this selective recurrence in two roles — a causal scan over time and, optionally, a bidirectional scan over variates — detailed next.

4. Method

4.1. Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [2, 11] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s \mathbf{X}_s + \mathbf{W}_t \mathbf{X}_t \in \mathbb{R}^{H \times C}, \qquad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H \times L}, \tag{3}$$

i.e., a DLinear-style map [11] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [6] (Fig. 2a). Instantiating the selective recurrence of Section 3.2, the per-step discrete parameters are $\bar{\mathbf{A}}_k = \exp(\mathbf{\Delta}_k \mathbf{A})$ and $\bar{\mathbf{B}}_k = \mathbf{\Delta}_k \mathbf{B}_k$, and the layer runs the recurrence

$$\mathbf{h}_k = \bar{\mathbf{A}}_k \, \mathbf{h}_{k-1} + \bar{\mathbf{B}}_k \, u_k, \qquad y_k = \mathbf{C}_k^{\top} \mathbf{h}_k + D \, u_k, \tag{4}$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\mathbf{\Delta}_k \in \mathbb{R},\ \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps — the input-dependent selectivity of [6]. The last state, summed with a whole-window linear embedding $\mathbf{W}_e \tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [7]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C \times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \qquad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \tag{5}$$

Because ϕ_{time} is zero-initialized, the branch's output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express.

4.2. Frequency Branch: Spectral Filtering and a Dormant Spectral Pathway

The frequency branch applies the real FFT along time, $\mathbf{Z} = \text{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C\times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F\times F}$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$ along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\text{freq}} \in \mathbb{R}^{C\times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS-style spectral pathway (optional, per dataset). A complex linear map $\mathbf{U} \in \mathbb{C}^{F'\times F}$ with $F' = \lfloor (L+H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C\times F'}$), maps the (low-passed) window spectrum to that of a length-$(L+H)$ series, whose inverse transform's last H samples are the branch's linear spectral forecast:

$$\mathbf{Y}_{\text{freq}} = \underbrace{\text{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[\,L{:}\,]}_{\text{linear-in-frequency}} + \phi_{\text{freq}}(\mathbf{f}_{\text{freq}}), \qquad \mathbf{U}, \phi_{\text{freq}} \equiv \mathbf{0} \text{ at init.} \tag{6}$$

This pathway uses the frequency interpolation idea of FITS [12], but is not a fitted FITS model: $\mathbf{U}$ is *zero-initialized*, so the frequency branch is silent at initialization (the length-L and length-$(L+H)$ frequency grids do not align, so no canonical identity init exists, and a random $\mathbf{U}$ would inject noise during the highest-learning-rate epochs). With the time head, spectral map, and

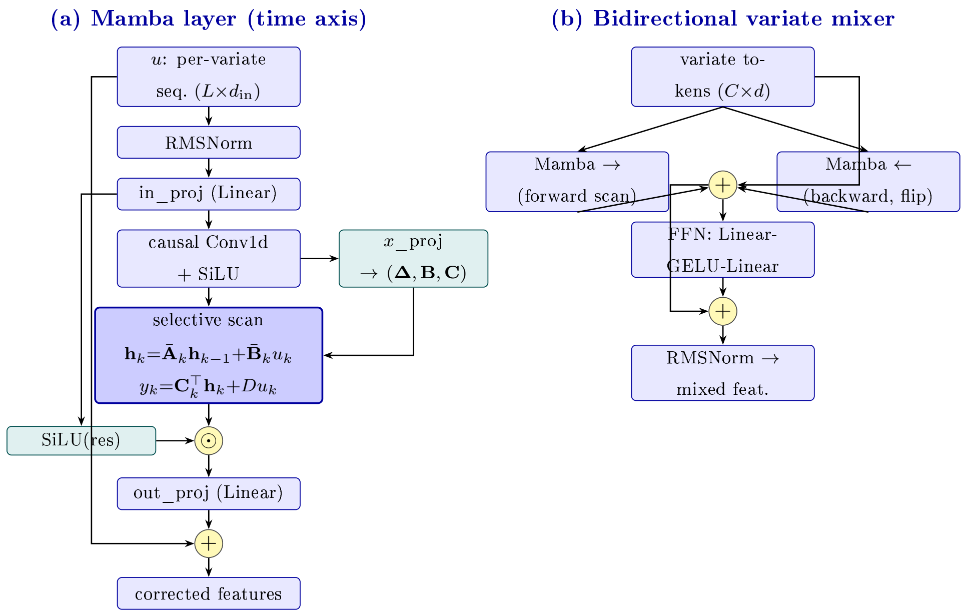

Figure 2: Internal structure of DD-Mamba's two selective-state-space blocks (cf. the blue boxes in Fig. 1). **(a)** The causal Mamba layer used as the time-axis encoder (Eq. (4)): an input-dependent selective scan, gated by a SiLU branch and wrapped in a residual, with $(\mathbf{\Delta}, \mathbf{B}, \mathbf{C})$ predicted per step from the input. **(b)** The bidirectional mixer used over the *variate* dimension (Section 4.3): forward and backward scans of the C channel tokens summed in one residual stream, then a position-wise feed-forward block (the S-Mamba design [7]).

gate all zero or constant at init, the end-to-end map in the normalized space is the linear $g_0 \cdot \mathbf{Y}_{\text{lin}}$; composed with RevIN this is the RLinear-class structure [15, 16] — linear in the normalized window, by construction not fitted. The spectral map is enabled per dataset (Table 2); where disabled, the frequency branch remains active through its encoder and forecast head.

4.3. Cross-Channel Mixing over Variates

Channel independence is a strong baseline [4, 11] but leaves accuracy on the table when variates are coupled [5, 7, 14]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the C per-channel features, followed by a position-wise feed-forward block (Fig. 2b). Bidirectionality is essential: channel order carries no causal direction, so each layer sums a forward and a reversed scan in one residual stream. Whether such mixing *should* pay is a property of the data: the mean absolute pairwise Pearson correlation (PCC) between channels on the training split spans 0.31–0.35 on the weakly coupled ETT family but 0.50–0.91 on Electricity, Traffic, and Solar (Table 1). We therefore expose the mixer as a per-dataset switch — on for strongly coupled, many-channel benchmarks, off for weakly coupled, few-channel ones (per-dataset settings in Section 5.1) — and test the decision in both directions in Section 5.3.

4.4. Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\text{time}} + (1-g) \odot \mathbf{Y}_{\text{freq}}, \qquad g = \sigma\big(\psi([\mathbf{f}_{\text{time}}; \mathbf{f}_{\text{freq}}])\big) \in (0,1)^{C\times 1}, \quad (7)$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value reports each branch's convex mixing weight — a lightweight proxy for its contribution, not a causal attribution. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with its bias set so that training starts gate-open toward the time branch ($g_0 \approx 0.9$; exact value in Section 5.1): the time branch is linear at initialization while the frequency branch is silent, and an even initial mix would halve the initial linear forecast. This is an initialization convention; empirical gate distributions are not available in the present study.

4.5. Training

We minimize MSE with AdamW under a halving schedule ($\eta_e = \eta_0 \cdot 2^{-(e-1)}$). All runs use 10 epochs with best-checkpoint early stopping. This schedule was held fixed across our runs but was not compared with cosine, warm-up, or longer-budget alternatives; schedule sensitivity therefore remains a potential confound.

5. Experiments

5.1. Setup

5.1.1. Datasets and Protocol

We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [5, 7]: look-back 96, horizons 96/192/336/720. ETT uses the canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

5.1.2. Baselines

We compare against the strongest published results under this protocol: S-Mamba [7], iTransformer [5], PatchTST [4], DLinear [11], and TimesNet

Table 1: Benchmark datasets, their cross-channel coupling, and where DD-Mamba places its capacity. $\overline{|r|}$: mean absolute pairwise Pearson correlation between channels on the training split (— = raw file unavailable in our build environment).

Dataset	Variates	Steps	Granularity	$\overline{|r|}$	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	0.31/0.35	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	0.31/0.35	off	0.24–0.36M
Weather	21	52,696	10 min	—	M=2, d=256	5.8M
Solar-Energy	137	52,560	10 min	0.91	M=1, d=128	0.96M
Electricity	321	26,304	1 hour	0.50	M=2, d=512	19.7M
Traffic	862	17,544	1 hour	0.58	M=2, d=512	19.7M
Exchange	8	7,588	1 day	0.48	off	0.09M

[17]. Numbers are quoted from [7] rather than reproduced: the protocol is identical, but training pipelines, seeds, and implementations differ across works, so gaps below the descriptive tolerance in Section 5.1.3 relative to quoted numbers should be read with caution (flagged in the text). We stress that, because the baselines are not rerun in a single pipeline, this comparison is *indicative, not confirmatory*, and we make no state-of-the-art claim on its basis. A unified rerun — of these baselines together with the recent time–frequency and Mamba models most relevant to our positioning (TF4TF [9], ms-Mamba [8]) and the strong linear map RLinear [15], under one implementation, data pipeline, seed set, and tuning budget — is the primary experiment needed to turn the comparison into evidence, and we flag it as a prerequisite for any accuracy claim.

5.1.3. Implementation and Reproducibility.

All experiments use PyTorch 2.11 (CUDA 13.0) on a single NVIDIA RTX 5090. Optimization is AdamW (grad clip 5.0), 10 epochs with the learning rate halved each epoch, early stopping on validation loss, and mixed precision with the selective-scan recurrence kept in full precision for numerical stability of the L-step state accumulation. The decomposition kernel is 25; Mamba blocks use expansion 2 and conv width 4 throughout; the fusion gate's bias is initialized to 2.2, giving $g_0 = \sigma(2.2) \approx 0.9$ (Section 4.4). Table 2 lists all per-dataset settings; configs and the ablation harness ship with the code. All main results are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003. The table records final empirical configurations, not a transferable capacity-selection rule: no common pre-registered search space or equalized tuning budget was used across all datasets. Consequently, capacity placement is reported as an empirical finding, and new datasets require validation-only selection of these switches.

5.1.4. Statistical Methodology

We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we form the *paired* per-seed difference and report its mean effect, a two-sided t-based p-value, and a 95% CI (n=3, t_{crit}=4.303). Because we run many such tests, raw p-values overstate significance, so we additionally control the false discovery rate with the Benjamini–Hochberg (BH) procedure *within each family* — the 29 component-placement comparisons and the 54 branch-removal comparisons — and report the resulting q-values (`scripts/compute_stats_correction.py`). An effect is called *confirmatory* only if it survives BH at $q<0.05$; raw-significant effects that do not survive are reported as *exploratory*. We flag $n=3$ as a real limitation of power: five and ten seeds are the natural strengthening, and effect sizes with CIs are reported throughout so that readers can judge magnitude independently of the significance verdict. *Against quoted baselines*, pairing is impossible, so 0.005 MSE is used only as a descriptive tolerance based on the largest standard deviation among our runs, not as a significance test. Our in-framework DLinear re-run shows why: through the identical pipeline on ETTh1 it averages 0.462/0.456 MSE/MAE versus its quoted 0.456/0.452 — a +0.006 implementation gap, the order of several margins in Table 3. Cross-paper comparisons therefore cannot support sub-0.01 distinctions; DD-Mamba's margin over the in-framework DLinear (0.439 vs. 0.462) is, by contrast, a same-pipeline comparison.

5.2. Main Results

Against the quoted results and under the stated descriptive tolerance, DD-Mamba shows average-MSE gaps larger than 0.005 on Solar, ETTh1, and ETTh2 (Table 3; per-horizon values in Table 4), although only the Solar (0.034) and ETTh1 (0.015) margins also exceed the wider 0.01 cross-paper caution of Section 5.1.3; ETTh2's 0.007 lies between the two thresholds. The largest margin is on Solar (0.199 vs. iTransformer's 0.233; −17% vs. S-Mamba, with the normalization-free configuration of Section 5.3). On ETTm1 and ETTm2 it improves over the quoted S-Mamba result but ties PatchTST, and on Weather and Electricity the gap is below the tolerance. It trails DLinear on Exchange (by 0.009, above tolerance) and S-Mamba on Traffic (by 0.024), which we *hypothesize* reflects the weekly cycle exceeding

Table 2: Per-dataset hyperparameters. d: model width; time enc.: time-branch encoder (Mamba layers in parentheses); M: variate-mixer layers; N: SSM state size; ρ: spectral low-pass fraction (Section 4.2); spec. map: the *optional* FITS-style pathway of Eq. (6) — when "no", the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. †Solar adopts IN = no from the ablation finding (Section 5.3); its main results were re-measured under this configuration across all horizons and seeds.

Dataset	d	time enc.	M	N	ρ	spec. map	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5{\cdot}10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5{\cdot}10^{-4}$	32	$0.3/5{\cdot}10^{-4}$
ETTm1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTm2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2{\cdot}10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2{\cdot}10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no†	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no	yes	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5{\cdot}10^{-4}$

the look-back. As all comparisons are to numbers *reported* in [7], we phrase the claim conservatively: DD-Mamba is competitive with results reported under this protocol, its clearest margins on Solar and ETTh1, where the linear backbone and capacity restraint matter most and DD-Mamba uses $\approx$ 0.25M parameters — roughly an order of magnitude fewer than variate-attention models configured for these datasets.

5.3. Component Ablations

Our ablation harness flips exactly one switch of a dataset's full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. We report its complete output on three datasets under their *final released*

Table 3: Long-term forecasting results averaged over horizons $H \in \{96, 192, 336, 720\}$ with look-back L=96 (MSE/MAE; lower is better). DD-Mamba entries are means over three seeds (Table 4 reports per-horizon std). Baselines quoted from [7]. **Bold** marks the best result *and* every result within the descriptive tolerance (< 0.005, Section 5.1.3) of it; this formatting is not a cross-paper significance claim; underline marks the next-best result.

Dataset	DD-Mamba (ours)	S-Mamba	iTransformer	PatchTST	DLinear	TimesNet
ETTh1	**0.439**/**0.434**	0.455/0.450	0.454/0.447	0.469/0.454	0.456/0.452	0.458/0.450
ETTh2	**0.374**/**0.401**	0.381/**0.405**	0.383/0.407	0.387/0.407	0.559/0.515	0.414/0.427
ETTm1	**0.386**/**0.398**	0.398/0.405	0.407/0.410	**0.387**/**0.400**	0.403/0.407	0.400/0.406
ETTm2	**0.280**/**0.325**	0.288/0.332	0.288/0.332	**0.281**/**0.326**	0.350/0.401	0.291/0.333
Weather	**0.248**/**0.276**	**0.251**/**0.276**	0.258/**0.278**	0.259/0.281	0.265/0.317	0.259/0.287
Electricity	**0.168**/**0.263**	**0.170**/**0.265**	0.178/0.270	0.205/0.290	0.212/0.300	0.192/0.295
Traffic	0.438/0.281	**0.414**/**0.276**	0.428/0.282	0.481/0.304	0.625/0.383	0.620/0.336
Exchange	0.363/**0.405**	0.367/0.408	0.360/**0.403**	0.367/**0.404**	**0.354**/0.414	0.416/0.443
Solar	**0.199**/**0.260**	0.240/0.273	0.233/**0.262**	0.270/0.307	0.330/0.401	0.301/0.319

configurations, three seeds each at H=96: ETTh1 and Solar (Table 5) and Weather (Table 7), with the paired methodology of Section 5.1.3. Two consistency checks pass: the Solar "full" row reproduces the main-table Solar H=96 entry ($0.180_{\pm.004}$), and its w/o-instance-norm variant coincides with the full (RevIN-off) configuration, so that cell is omitted.

Component-placement effects are directional but exploratory.. A two-sided pattern spans datasets at the raw level: adding the variate mixer on ETTh1 *hurts* (by +0.0077 MSE, raw p=0.014) while removing it on Weather also hurts (by +0.0104, raw p=0.022); normalization shows the same shape (removing it hurts Weather +0.0161, raw p=0.023, while disabling it on Solar improved the re-measured main results from 0.228 to 0.199). We emphasize, however, that across the 29 component-placement comparisons *none* survives Benjamini–Hochberg correction (the five raw-significant effects have

Table 4: DD-Mamba per-horizon results (mean ± std over three seeds), look-back L=96.

Dataset		H=96	H=192	H=336	H=720	Avg
ETTh1	MSE	0.380$\pm_{.001}$	0.431$\pm_{.002}$	0.474$\pm_{.004}$	0.470$\pm_{.003}$	0.439
	MAE	0.396$\pm_{.001}$	0.424$\pm_{.001}$	0.447$\pm_{.003}$	0.468$\pm_{.001}$	0.434
ETTh2	MSE	0.293$\pm_{.001}$	0.368$\pm_{.003}$	0.414$\pm_{.001}$	0.421$\pm_{.001}$	0.374
	MAE	0.344$\pm_{.002}$	0.391$\pm_{.000}$	0.426$\pm_{.000}$	0.441$\pm_{.001}$	0.401
ETTm1	MSE	0.320$\pm_{.001}$	0.366$\pm_{.002}$	0.398$\pm_{.001}$	0.462$\pm_{.001}$	0.386
	MAE	0.359$\pm_{.001}$	0.384$\pm_{.001}$	0.406$\pm_{.001}$	0.443$\pm_{.001}$	0.398
ETTm2	MSE	0.177$\pm_{.000}$	0.242$\pm_{.002}$	0.302$\pm_{.000}$	0.401$\pm_{.003}$	0.280
	MAE	0.260$\pm_{.001}$	0.303$\pm_{.001}$	0.340$\pm_{.000}$	0.398$\pm_{.002}$	0.325
Weather	MSE	0.162$\pm_{.000}$	0.212$\pm_{.000}$	0.269$\pm_{.001}$	0.352$\pm_{.003}$	0.248
	MAE	0.207$\pm_{.001}$	0.255$\pm_{.000}$	0.295$\pm_{.002}$	0.349$\pm_{.002}$	0.276
Electricity	MSE	0.140$\pm_{.000}$	0.157$\pm_{.001}$	0.174$\pm_{.001}$	0.200$\pm_{.004}$	0.168
	MAE	0.235$\pm_{.000}$	0.252$\pm_{.001}$	0.271$\pm_{.001}$	0.296$\pm_{.004}$	0.263
Traffic	MSE	0.402$\pm_{.002}$	0.424$\pm_{.001}$	0.441$\pm_{.001}$	0.485$\pm_{.003}$	0.438
	MAE	0.266$\pm_{.001}$	0.275$\pm_{.001}$	0.283$\pm_{.000}$	0.301$\pm_{.000}$	0.281
Exchange	MSE	0.086$\pm_{.000}$	0.179$\pm_{.001}$	0.330$\pm_{.002}$	0.857$\pm_{.001}$	0.363
	MAE	0.205$\pm_{.000}$	0.301$\pm_{.000}$	0.416$\pm_{.002}$	0.699$\pm_{.000}$	0.405
Solar	MSE	0.180$\pm_{.004}$	0.195$\pm_{.005}$	0.207$\pm_{.001}$	0.212$\pm_{.005}$	0.199
	MAE	0.237$\pm_{.003}$	0.259$\pm_{.003}$	0.270$\pm_{.001}$	0.274$\pm_{.005}$	0.260

$q \geq 0.11$): at three seeds these are *exploratory, directional* findings, not confirmed effects, and we report them as such. The consistency of the *direction* (component value is dataset-dependent, in both directions) is what motivates per-dataset switches; confirming any individual effect requires more seeds. Zero initialization is a *null* result — no significant difference on either dataset (−0.0012 ETTh1, −0.0046 Solar; CIs include zero) — so we make no accuracy claim for it, only the interpretable linear start (Section 4.1). The DLinear backbone and the learned gate versus averaging are within CIs everywhere; on Weather (Table 7) three components have unadjusted paired CIs excluding zero (normalization, the variate mixer, and the temporal Mamba,

Table 5: Component ablations on ETTh1 and Solar-Energy (H=96, three seeds, final configurations): paired per-seed ΔMSE to the full model with t-based 95% CIs; * CI excludes zero. The spectral map is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802±.001	0.1802±.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
spectral map removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

+0.0027, positive on all seeds). The recurring pattern — component value is dataset-dependent, in *both* directions — is the argument for per-dataset switches over universal defaults, and it does not depend on any single dataset.

Branch matrix: where each domain matters.. Because a single Solar cell cannot carry a dual-domain conclusion, we ran the full branch-ablation matrix — full / time-only / freq-only under paired seeds on all nine datasets at H=96 and across all four horizons on the six ETT/Weather/Solar datasets, 27 cells in total (Table 6). Every full-model mean reproduces the corresponding main-table entry, and the Solar H=96 cell replicates the earlier ablation (+0.0126 vs. +0.0114, overlapping CIs). The 27 cells yield 54 branch-removal tests; we report both raw significance (CI excludes zero, marked * in Table 6) and BH-corrected q-values, and treat only the latter as confirmatory.

Frequency branch. At the raw level, removing it is significant in 5 cells, concentrated at the shortest horizon: Solar (+0.0126), ETTm1 (+0.0073), Electricity (+0.0019), and Exchange (+0.0015) at H=96, plus ETTm2 at H=336. After correction, however, only **Solar at** H=96 survives (q=0.048); the other four fall to $q \geq 0.10$. So confirmatory evidence for the frequency branch is limited to a single dataset and horizon, and its apparent breadth across four benchmarks is *exploratory*.

Time branch. Removing it is significant in 14 cells raw and **9 survive BH** (q<0.05): ETTm1 at *every* horizon (+0.017–+0.021, the most persistent effect we measure), ETTm2 at H=96 and H=336, and Electricity and Exchange at H=96. The time branch is thus robustly necessary on the ETTm family; on ETTh1, ETTh2, and Weather no branch effect is detected at *any* horizon, nor on Traffic at H=96.

The dual-domain benefit is therefore real but sharply conditional — on the dataset *and* the horizon — and, after multiplicity control, its confirmed frequency-side evidence rests on Solar@96 alone, with the time branch carrying the robust component signal. Elsewhere the unused path is a no-cost redundancy behind the convex gate.

Negative results that shaped the design.. Deepening the traffic mixer from two to four layers (19.7 $\rightarrow$ 37.9M) gave no gain, so the release keeps two; enabling the mixer on ETT (8.4K training windows) tripled parameters and was associated with premature early stopping and worse held-out error, so the released ETT configurations disable it.

Table 6: Branch-ablation matrix (27 cells): paired per-seed ΔMSE versus the full model when one branch is removed (three seeds; * marks *raw* t-based 95% CI excluding zero; per-cell CIs ship with the results). After Benjamini–Hochberg correction over the 54 tests, 9 survive at $q<0.05$ — eight on the time side (ETTm1 all horizons; ETTm2@96/336; Electricity@96; Exchange@96) and, on the frequency side, *only* Solar@96 (q=0.048); the remaining starred cells are exploratory. Electricity, Traffic, and Exchange were run at H=96 only (—). Full-model means match Table 4.

	freq removed: ΔMSE at H				time removed: ΔMSE at H			
Dataset	96	192	336	720	96	192	336	720
ETTh1	+.0034	+.0005	−.0047	+.0096	−.0007	−.0001	−.0003	+.0176
ETTh2	+.0006	+.0078	+.0054	+.0093	−.0031	−.0017	−.0017	+.0026
ETTm1	+.0073*	+.0002	+.0019	−.0005	+.0196*	+.0168*	+.0203*	+.0210*
ETTm2	+.0020	+.0014	+.0030*	+.0001	+.0060*	+.0060	+.0059*	+.0059
Weather	+.0008	+.0004	−.0002	−.0009	+.0006	+.0011	+.0021	+.0027
Solar	+.0126*	−.0038	+.0006	−.0032	+.0052	+.0075	+.0040*	+.0027
Electricity	+.0019*	—	—	—	+.0017*	—	—	—
Traffic	−.0015	—	—	—	+.0013	—	—	—
Exchange	+.0015*	—	—	—	+.0058*	—	—	—

5.4. Analysis

Where each domain wins.. The matrix (Table 6) separates three regimes, with the caveat that only the BH-surviving cells are confirmatory. Solar is the one dataset where the frequency path is confirmed load-bearing at H=96 (+0.0126, q=0.048; removing the time branch is within noise), and even there the benefit vanishes at longer horizons — consistent with Solar's 10-minute daily cycle spanning $144>L=96$ samples, so the spectral path cannot observe a complete daily period and its measured advantage comes from shorter-scale detail that matters most for near-term steps. On ETTm1, Electricity, and Exchange the branches *appear* complementary at H=96, but

Table 7: Component ablation on Weather (H=96; MSE mean ± std over three seeds). Δ is the *paired* per-seed difference to the full model with a t-based 95% confidence interval (n=3); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620±.0003	—	reference
w/o instance norm	0.1781±.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724±.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647±.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631±.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628±.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o spectral map	0.1627±.0016	+0.0007 [−0.0041, +0.0055]	FITS-style pathway
freq branch only	0.1626±.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626±.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622±.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

after correction it is the *time* branch that is confirmed (the frequency side is exploratory there), and on ETTm1 the time branch stays confirmed at every horizon. On ETTh1, ETTh2, and Weather (all four horizons) and Traffic the branches are interchangeable, and the convex gate makes carrying the unused path harmless. Which domain a dataset needs — and for how far ahead — is itself dataset-dependent; but the strongest, correction-surviving statement the data support is that the *time* branch is broadly necessary while a confirmed *frequency* contribution is specific to Solar at short horizons.

Data correlations predict — and mislead about — the mixer.. A data-mining reading of the placement finding asks *which measurable property of a dataset* decides whether cross-variate capacity pays. Mean absolute cross-channel PCC (Table 1) separates the two regimes almost perfectly: the mixer-off ETT family sits at $\overline{|r|} = 0.31$–0.35 (only 14–24% of channel pairs exceed $|r|>0.6$), while the mixer-on Electricity, Traffic, and Solar span 0.50–0.91 (Solar's near-total 0.91 reflects neighboring plants sharing irradiance), matching where the paired ablations find the mixer load-bearing versus harmful. Exchange is the instructive exception: $\overline{|r|} = 0.48$, yet the mixer is disabled — its currency series are near random walks whose correlation is co-trending rather than exploitable cross-sectional signal, and with only 7.6K steps the mixer's parameters cannot be estimated reliably. Raw correlation is therefore a useful *screen*, not a decision rule: high PCC flags candidates for cross-variate capacity, but non-stationarity and sample size can inflate it, and the seed-paired ablation remains the arbiter.

Complexity accounting.. Table 8 reports model-only, hardware-independent estimates rather than comparative runtime evidence. Parameter count is governed by model *width*, not channel count (electricity and traffic share 19.7M despite a 2.7× gap in variates), so model size is decoupled from data dimensionality; compute and activation memory instead scale with the channel count through the variate mixer (traffic 60 GMACs vs. electricity's 22 at equal width, while ETT and exchange sit two orders of magnitude below), keeping the capacity restraint on those datasets inexpensive. Wall-clock throughput and matched-baseline runtime remain unmeasured.

6. Discussion

6.1. Practical guidance: which components for which data?

The placement evidence condenses into a usable screen. *(i) Variate mixer:* compute the mean absolute cross-channel PCC on the training split; values

Table 8: Model-only complexity profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply–accumulates per forecast (`Linear` + `Conv1d` + the selective-scan recurrence); act. mem.: forward activation footprint (fp32). All three columns are hardware-independent estimates; no wall-clock or matched-baseline runtime is claimed.

Dataset	Variates	Params (M)	GMACs	Act. mem. (MB)
Exchange	8	0.09	0.02	2.3
ETTh1	7	0.25	0.08	4.0
Weather	21	5.77	1.90	46.6
Solar	137	0.96	3.26	150.4
Electricity	321	19.70	22.4	254.3
Traffic	862	19.70	60.2	682.7

near or above 0.5 with many channels flag candidates for cross-variate capacity (Electricity, Traffic, Solar), while weakly coupled, few-channel data (the ETT family) should leave it off — and co-trending, non-stationary series (Exchange) can inflate PCC without exploitable structure, so a seed-paired validation ablation should confirm the choice. *(ii) Frequency branch:* its measured value concentrates at short horizons and on strongly periodic data; at long horizons the shape-level information dominates and the branch becomes a no-cost redundancy behind the convex gate. *(iii) Normalization:* RevIN helps where windows carry genuine distribution shift and hurts where apparent nonstationarity is itself periodic structure (Solar's night zeros), so it, too, should be validated per dataset rather than defaulted.

6.2. Dispersion diagnostics: why normalization is dataset-dependent

A training-free diagnosis explains the two-sided RevIN result (Fig. 3). Standardizing each channel and taking a rolling standard deviation over the dominant period, all three probed datasets carry substantial time-varying scale (coefficient of variation of the rolling std 0.20–0.48); a KPSS test rejects stationarity of the rolling-std series in every sampled channel, and a Ljung-Box test finds it strongly autocorrelated. The scale is therefore non-stationary and predictable in *all* cases — yet normalizing by the input-window estimate helps Traffic (+0.100 MSE when removed) and hurts Solar (−0.029 when removed): on Solar the window statistics mostly encode the night fraction of the window, and de-normalization amplifies errors by an unstable standard deviation. The distinction is structural, not statistical, which is why we ship normalization as a verified per-dataset switch. Forecasting the predictable component of the scale explicitly — an STD-style dispersion path [25] — is a natural extension that we prototype in the released code but do not claim as part of the evaluated model.

6.3. Threats to validity

Several threats remain, each with the experiment that would resolve it.

Quoted baselines.. All baselines except the ETTh1 DLinear re-run are quoted from [7] rather than reproduced in one pipeline, so no accuracy claim should be read as confirmatory until a unified rerun (adding TF4TF [9], ms-Mamba [8], and RLinear [15]) is completed.

Statistical power and multiplicity.. With n=3 seeds, tests have limited power, and although we apply Benjamini–Hochberg correction, all 29 component-

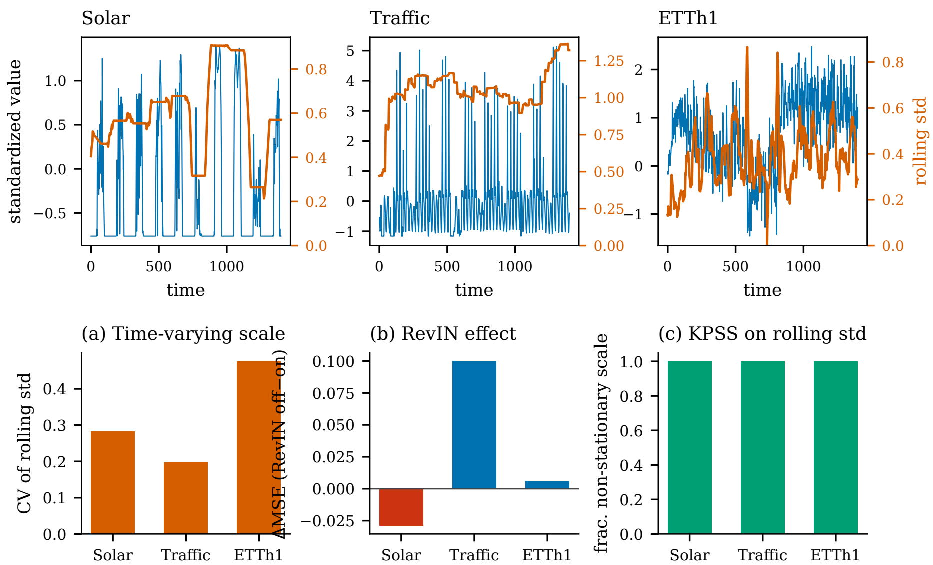

Figure 3: Training-free dispersion diagnostics. *Top:* a representative standardized channel (blue) with its rolling standard deviation (orange) for Solar, Traffic, and ETTh1. **(a)** Coefficient of variation of the rolling std. **(b)** Measured RevIN effect (ΔMSE, off$-$on): normalizing by the input scale helps Traffic but hurts Solar. **(c)** Fraction of sampled channels whose rolling-std series a KPSS test declares non-stationary.

placement effects fall to $q \geq 0.11$: those results are exploratory. Raising to five and preferably ten seeds, and reporting raw p- and corrected q-values for every comparison (as we do in `stats_correction.json`), is needed to move the placement effects from directional to confirmatory.

Model-selection bias.. This is the most consequential threat. Although checkpoints are chosen on validation loss, some *switch* settings — most visibly disabling RevIN for Solar — were decided after inspecting test-set ablations and then folded into the reported configuration. This is not training-data leakage, but it is selection on the test signal, and it can inflate the reported numbers for the affected datasets. The clean protocol, which we recommend for the revision, fixes a candidate configuration space in advance, selects *every* switch (normalization, mixer, spectral map, encoder type) on validation data only, freezes the choice before touching the test set, and reports three clearly labeled models: a single *fixed* configuration applied everywhere, a *validation-selected* model, and a *test-oracle* upper bound. Adding untouched confirmatory datasets (e.g., PEMS03/04/07/08) or previously unused temporal test blocks would then verify that the validation-only selection logic generalizes rather than having been fit to the existing nine benchmarks.

Coverage and efficiency.. Electricity, Traffic, and Exchange enter the branch matrix at H=96 only, and the other component ablations cover only H=96; schedule sensitivity and head-norm dynamics are not measured; and the complexity table lacks wall-clock, matched-baseline throughput, latency, and peak-memory figures. The fixed 96-step look-back leaves the Traffic weekly-cycle hypothesis untested, while Exchange at H=720 remains a notable failure case. Explicit forecasting of time-varying dispersion is left as future work rather than claimed as part of the evaluated model.

7. Conclusion

DD-Mamba is a forecast-decomposable framework built on state-space branches — a DLinear-style temporal backbone, a selective-state-space correction, an optional frequency pathway, and a switchable cross-variate mixer, each an independently removable component — whose purpose is to diagnose *when* such components pay rather than to claim universal superiority. Against quoted results under the look-back-96 protocol it is competitive, with its clearest margins on Solar and ETTh1; because the baselines are quoted rather than rerun in one pipeline, we draw no state-of-the-art conclusion from the comparison. The framework's value is the multiplicity-controlled diagnosis it enables. Under Benjamini–Hochberg correction, all 29 component-placement effects are exploratory at three seeds (directionally consistent but not confirmed), while the branch matrix confirms 9 of 54 tests: the *time* branch is robustly necessary on the ETTm datasets, whereas confirmatory evidence for the *frequency* branch narrows to Solar at H=96. Zero initialization is accuracy-neutral and retained only for its interpretable linear starting point. The honest reading is thus conditional component effectiveness: a confirmed, dataset-specific frequency benefit on Solar, a broadly confirmed time branch, and a set of directional placement effects that a higher-powered study should adjudicate. Strengthening the protocol — unified baseline reruns, validation-only switch selection with a labeled test-oracle bound, more seeds, confirmatory datasets, and matched efficiency measurements — is the path from this diagnostic framework to a submission with confirmatory accuracy claims.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Second Author:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All nine benchmark datasets are publicly available (ETT, Weather, Solar, Electricity, Traffic, Exchange). An anonymized reproducibility package is provided *at submission* (not withheld until acceptance): it contains data-download scripts, exact environment versions and backend information, per-dataset configuration snapshots, commit identifiers, the ablation and branch-matrix runners, the multiplicity-correction script and its per-comparison p- and q-value outputs, seed-level result files, training logs, and the table-generation scripts; model checkpoints are included where size permits.

Declaration of generative AI in the writing process

During the preparation of this work the authors used generative AI tools to assist with language editing and code scaffolding; the authors reviewed and edited all content and take full responsibility for it.

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